

PPO in practice - reward shaping:

<< The 37 implementation details of PPO >>

PPO is idealization for language modeling.

PPO outer loop. Invoke an inner loop to optimize the loss over some rollouts.
Loss computation: AlpacaFarm code - loss computation, pretty standard stuff

PPO in conceptual level: the objective is basically all there is to it at a conceptual level

pseudo-codes for PPO-Clip algorithm

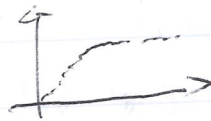
PPO in practice: generalized advantage estimate

instead of rewards, we use advantages

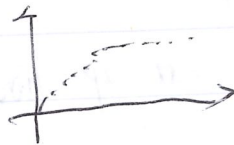
what do you expect to see in PPO?

reasonable training curves

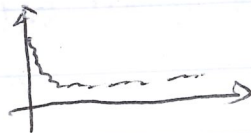
①: increasing overall rewards:



②: incline reward model:



③: Negative KL Rewards:



we still need another RL algorithm like: GRPO?

↳ why not PPO?

①: practically, complicated implementation

②: Value model (memory hungry, involves additional tuning for training)

↳ why not DPO?

①: Data is not inherently pairwise (form of

②: offline

Bradley-Terry comparison)

GRPO: group Relative policy optimization [shao et al; 2024]

①: start with PPO (similar structure)

②: Remove value function / advantage computation

③: calculate the advantage as "z-score within group"

workflow:

one prompt → multiple answers → ~~rank~~ compare answers in a group

reward better answers ← relative ranking ← importance / preference

⇒ PPO: ~~is~~ is this answer better than expected?

⇒ GRPO: is this answer better than the other answers?

↳ simple/elm implementation for GRPO:

①: compute Reward for each rollout

②: Mean/Var normalization per group

③: compute KL term / gradient updates on the loss